Magneto: Combining Small and Large Language Models for Schema Matching

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Biomedical Data Harmonization


Cell
CellPress
Volume 179, Issue 4, 31 October 2019, Pages 964-983.e31
Resource
Integrated Proteogenomic Characterization of Clear Cell Renal Cell Carcinoma

Cell
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Resource
Proteogenomic Landscape of Breast Cancer Tumorigenesis and Targeted Therapy
Karsten Krug, Eric J. Jaehnig, Shankha Satpathy, Lili Blumenberg, Alla Karpova,

Cell
CellPress
Volume 184, Issue 19, 16 September 2021, Pages 5031-5052.e26
Resource
Proteogenomic characterization of pancreatic ductal adenocarcinoma
Liwei Cao, Chen Huang, Daniel Cui Zhou, Yingwei Hu, T. Mamie Lih, Sara R. Savage, Karsten Krug, David J. Clark, Michael Schnaubelt, Lijun Chen, Felipe da Veiga Leprevost, Rodrigo Vargas Eguez, Weiming Yang, Jianbo Pan, Bo Wen, Yongchao Dou, Wen Jiang, Yuxing Liao, Zhiao Shi, Nadezhda V. Terekhanova, Grace Zhao

Biomedical Papers

+ Conduct meta-study
+ Publish research data to data commons

Data Commons

NIH NATIONAL CANCER INSTITUTE
Proteomic Data Commons

NIH NATIONAL CANCER INSTITUTE
GDC Data Portal

biobank uk
Enabling scientific discoveries that improve human health

Both labor-intensive and error-prone!

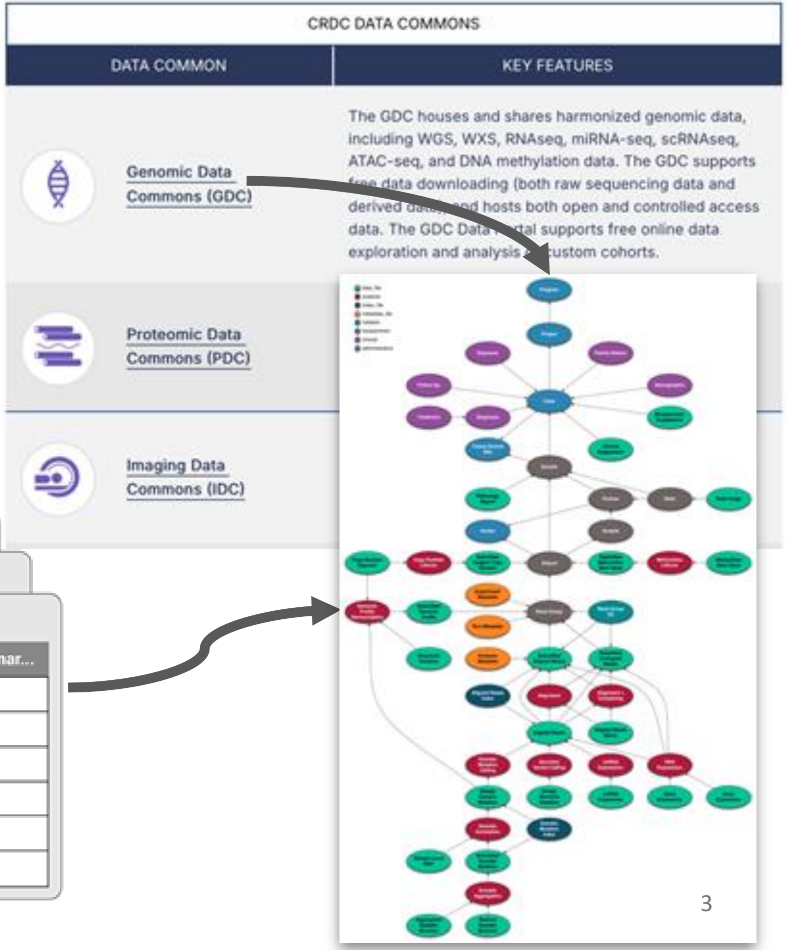
Biomedical Data Harmonization

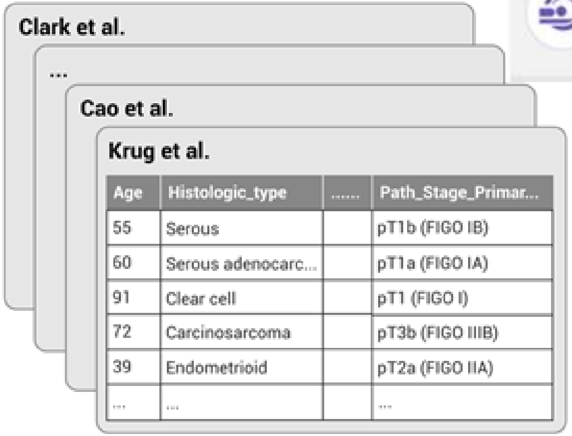
Motivation: Comprehensive proteogenomic analysis of different cancer types [Li et al. 2023]

Source: 10 datasets from different studies

Target: Genomic Data Commons (GDC) containing 700+ attributes

Goal: Map each dataset to a common schema and concatenate the harmonized data





Age	Histologic_type	......	Path_Stage_Primar...
55	Serous		pT1b (FIGO IB)
60	Serous adenocarc...		pT1a (FIGO IA)
91	Clear cell		pT1 (FIGO I)
72	Carcinosarcoma		pT3b (FIGO IIIB)
39	Endometrioid		pT2a (FIGO IIA)
...	...		...

Biomedical Data Harmonization

CRDC DATA COMMONS	
DATA COMMON	KEY FEATURES
Genomic Data Commons (GDC)	The GDC houses and shares harmonized genomic data, including WGS, WXS, RNAseq, miRNA-seq, scRNAseq, ATAC-seq, and DNA methylation data. The GDC supports free data downloading (both raw sequencing data and derived data) and hosts both open and controlled access data. The GDC Data Portal supports free online data exploration and analysis of custom cohorts.
Proteomic Data Commons (PDC)	
Imaging Data Commons (IDC)	

Clark et al.

...

Cao et al.

Krug et al.

Age	Histologic_type	......	Path_Stage_Primar...
55	Serous		pT1b (FIGO IB)
60	Serous adenocarc...		pT1a (FIGO IA)
91	Clear cell		pT1 (FIGO I)
72	Carcinosarcoma		pT3b (FIGO IIIB)
39	Endometrioid		pT2a (FIGO IIA)
...	...		...

Dataset (Concatenated)

Age	Histologic_type	......	Path_Stage_Primar...
55	Serous		pT1b (FIGO IB)
60	Serous adenocarc...		pT1a (FIGO IA)
91	Clear cell		pT1 (FIGO I)
72	Carcinosarcoma		pT3b (FIGO IIIB)
39	Endometrioid		pT2a (FIGO IIA)
73	Serous		pT1 (FIGO I)
41	Endometrioid		pT1a (FIGO IA)
93	Clear cell		pT1 (FIGO I)
77	Clear cell		pT3 (FIGO III)
80	Clear cell		pT2b (FIGO IIB)
39	Endometrioid		pT2a (FIGO IIA)
73	Serous		pT1 (FIGO I)
41	Endometrioid		pT1a (FIGO IA)
...	...		...

What Is Schema Matching?

Goal: Identify semantically equivalent columns across distinct data schemas

Input: Source table S, Target table T

Age	Histologic_type	Tumor_Stage-Pathological	…
35	Carcinosarcoma	Stage I	…
68	Clear Cell	Stage III	…

Source Table

age_at_diagnosis	ajcc_pathlogic_stage	Tumor_Focality	primary_diagnosis	uicc_pathlogic_stage	…
27153	Stage I	Multifocal	Abdominal desmoid	Stage I	…
30192	Stage IIIA	Unifocal	Carcinoma, NOS	Stage IIIA	…

Target Schema

What Is Schema Matching?

Goal: Identify semantically equivalent columns across distinct data schemas

Input: Source table S, Target table T

Output: Mapping $M \subseteq S \times T$ of semantically-equivalent columns

Age	**Histologic_type**	**Tumor_Stage-Pathological**	**…**
35	Carcinosarcoma	Stage I	…
68	Clear Cell	Stage III	…

age_at_diagnosis	**ajcc_pathlogic_stage**	**Tumor_Focality**	**primary_diagnosis**	**uicc_pathlogic_stage**	**…**
27153	Stage I	Multifocal	Abdominal desmoid	Stage I	…
30192	Stage IIIA	Unifocal	Carcinoma, NOS	Stage IIIA	…

Challenge: Nuanced semantics

Nuanced differences are common in biomedical data, making harmonization error-prone even for experts.

Age
28
59
76

age_at_diagnosis
18250
29220
21915

Age at the time of diagnosis in number of days since birth.

age_at_onset
55
72
91

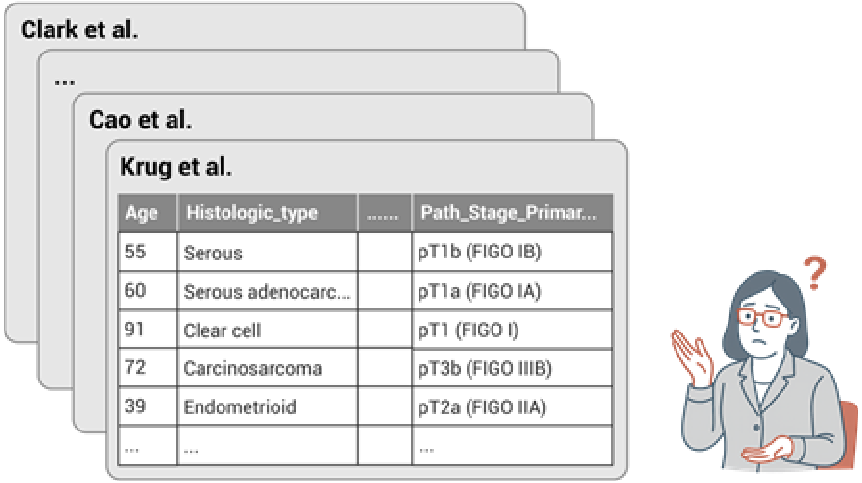
Numeric value used to represent the age of the patient when exposure to a specific environmental factor began.

age_at_index
60
75
58

The patient's age (in years) on the reference or anchor data used during date obfuscation.

Challenge: Biomedical Knowledge

Biomedical domain presents complex terminologies and it needs deep domain expertise.



Challenge: Scalability

In the biomedical domain, due to the large number of attributes in datasets and data commons, there are too many matches to verify.

Source Dataset

Age	Histologic_type	......	Path_Stage_Primar...
55	Serous		pT1b (FIGO IB)
60	Serous adenocarc...		pT1a (FIGO IA)
91	Clear cell		pT1 (FIGO I)
72	Carcinosarcoma		pT3b (FIGO IIIB)
39	Endometrioid		
...	...		

170 Attributes

Target Schema (Dataset)

NIH NATIONAL CANCER INSTITUTE Cancer Research Data Commons

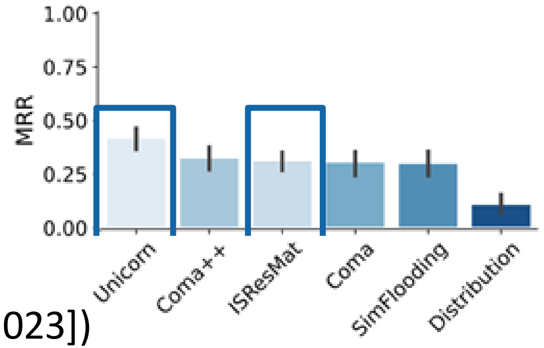
age_at_index	age_at_diagnosis	age_at_onset	......	attr_#734
67	29833	35		val
56	17990	66		val
31	8761	69		val
75	30101	73		val
78	26405	54		
...	...	...		

700+ Attributes

Landscape of Schema Matchers

***Goal:* Return *top-k ranked matches* for each source column**

- **Heuristic / Overlap-Based** (e.g., Jaccard, distribution-based [Zhang et al. 2011])
  - Miss semantic variation
- **Graph & Constraint-Based** (e.g., SimFlooding [Melnik et al. 2002])
  - Struggle with large, wide schemas
- **Ensembled Methods** (e.g., COMA [Do et al. 2002], COMA++)
  - Require manual tuning
- **Embedding-Based** (e.g., ISResMat [Du et al. 2024], Unicorn [Tu et al. 2023])
  - Still rely on fine-tuning and curated training data

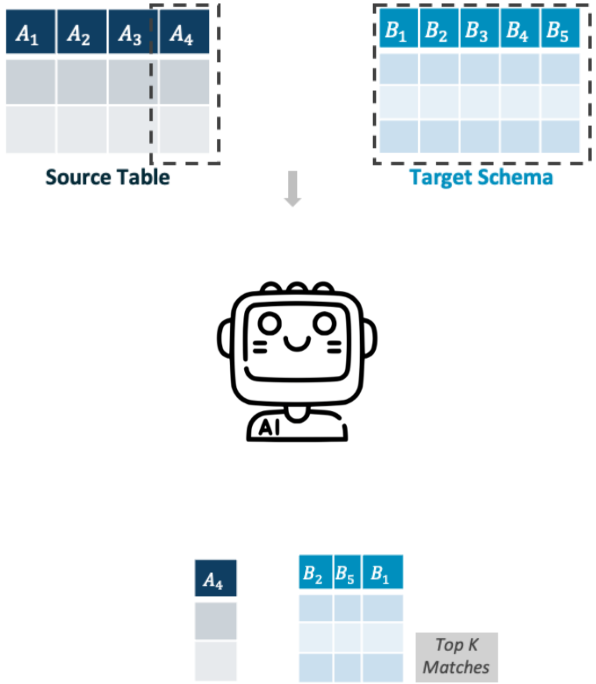


Lack of domain knowledge and semantic understanding

LLMs as Schema Matchers

+ Extensive world knowledge
+ Zero-shot generalization
+ Semantic depth

– Inference cost and latency
– Prone to errors with many candidates
– Token overhead with long context input



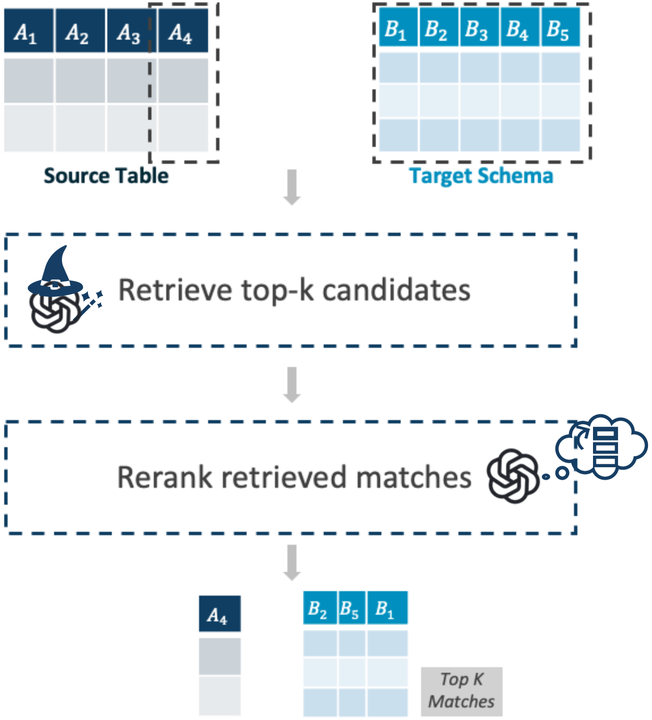
Magneto: A Retrieve-Rerank Framework

Goal: *Return* ***top-k ranked matches*** *for each source column*

- **Retrieval**
  Use Small Language Models (SLM) to efficiently retrieve top-k candidates for each source column

- **Reranking**
  Rerank retrieved candidates with specific criteria

- **Combining LLMs and SLMs**



Retrieval: Table Representation

How to represent tabular data with SLMs to obtain high-quality column embeddings?

- SLMs are trained for text, not tables
- Limited context window when columns contain many values

- **Column Encoding**
  - Use a pretrained SLM to convert columns into embeddings

Source Column

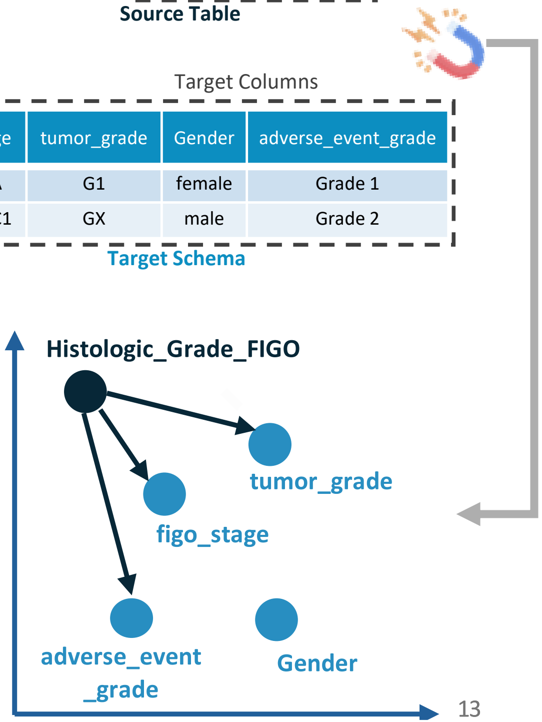
PT	Sex	Histologic_Grade_FIGO
T2	Female	FIGO grade 1
T4a	Male	FIGO grade 2
T3	Male	FIGO grade 1

Source Table

Target Columns

figo_stage	tumor_grade	Gender	adverse_event_grade
Stage IA	G1	female	Grade 1
Stage IIIC1	GX	male	Grade 2

Target Schema



Retrieval: Table Representation

- **Value Sampling**
  - Use priority sampling to balance frequent values with hash-based consistency
- **Column Serialization**
  - Introduce novel serialization strategies for different contexts: S_{default}, S_{verbose}, S_{repeat}
- **Column Encoding**
  - Use a pretrained SLM to convert columns into embeddings
- **K-Nearest Neighbor Search**
  - Retrieve top-k candidates (by cosine similarity)

Source Column

PT	Sex	Histologic_Grade_FIGO
T2	Female	FIGO grade 1
T4a	Male	FIGO grade 2
T3	Male	FIGO grade 1

Source Table

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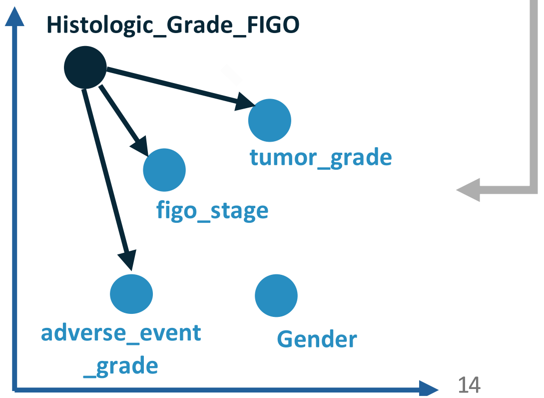


Challenge: SLMs Lack Domain Knowledge

- ❗ Zero-shot SLMs lack domain knowledge
  - ❗ Struggle with biomedical terminologies and contextual nuances
  - ❗ Fine tuning needs labeled training data

- Idea: **Distill LLM knowledge into SLMs**
  - LLMs capture rich cross-domain semantics but are computationally expensive
  - SLMs are efficient but need domain adaptation

Using LLM-Derived Data to Fine-Tune SLMs

- Use **LLMs** to **generate synthetic schema matching pairs** for each target column

tissue_type
Abnormal
Normal
Tumor
Peritumoral
Not Reported

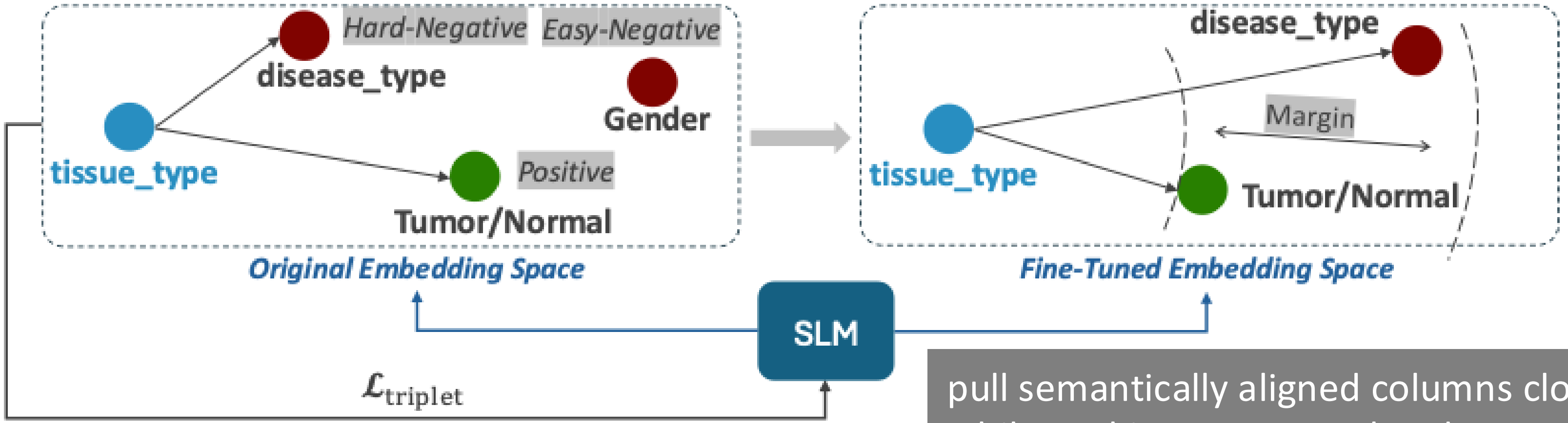
Target Column

sample_type	Tumor/Normal
Standard	Diseased
Collection_Not_Permitted	Cancerous
Malignant	Near_Tumor
Tumor_Border	Peritumoral
	Unreported

Synthetic Matches

Using LLM-Derived Data to Fine-Tune SLMs

- Use **LLMs** to **generate synthetic schema matching pairs** for each target columns
- Apply contrastive learning using triplet loss and online triplet mining to fine-tune SLMs on these LLM-generated pairs



pull semantically aligned columns closer while pushing apart unrelated ones

Using LLM-Derived Data to Fine-Tune SLMs

- Use **LLMs** to **generate synthetic schema matching pairs** for each target columns
- Apply contrastive learning using triplet loss and online triplet mining to fine-tune SLMs on these LLM-generated pairs

Key Advantages

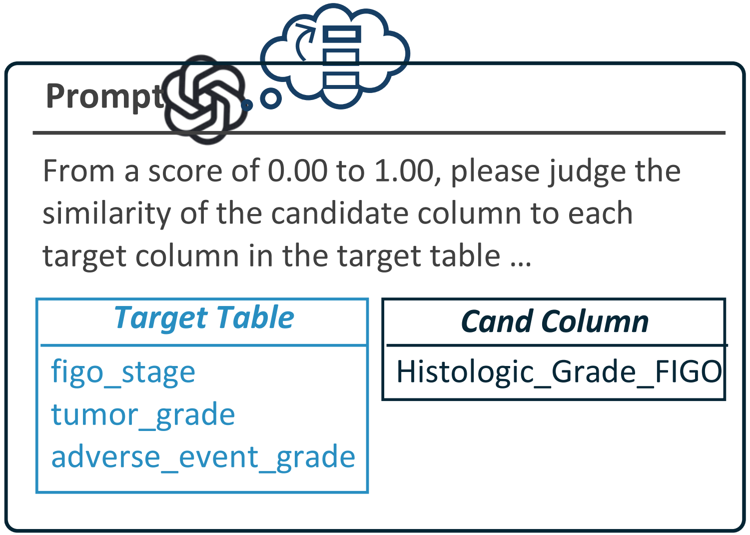
- One-time LLM calls during training only
- LLMs generate diverse domain examples
- SLMs inherit LLM knowledge but remain lightweight and fast at inference

LLM-Based Match Reranker

❗ Fine-tuned SLMs lack broad generalization

Idea: Reduced candidate pool enables affordable LLM reranking

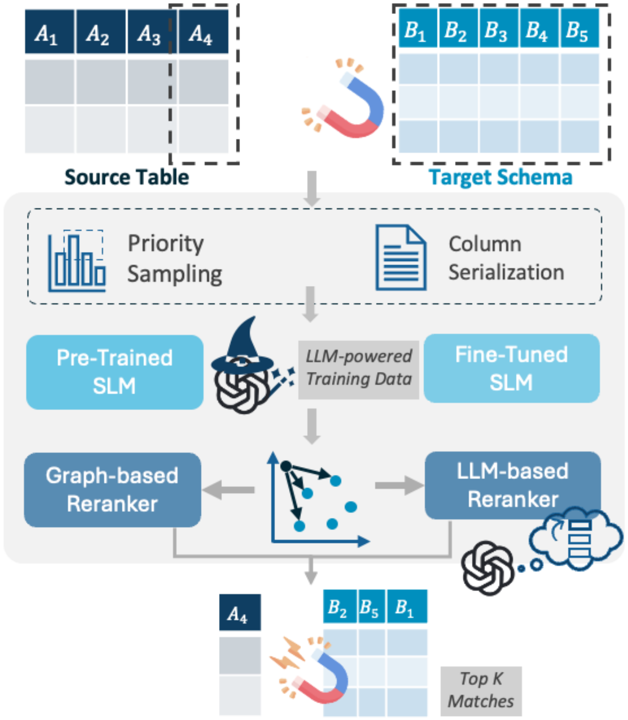
LLM-Based Match Reranker



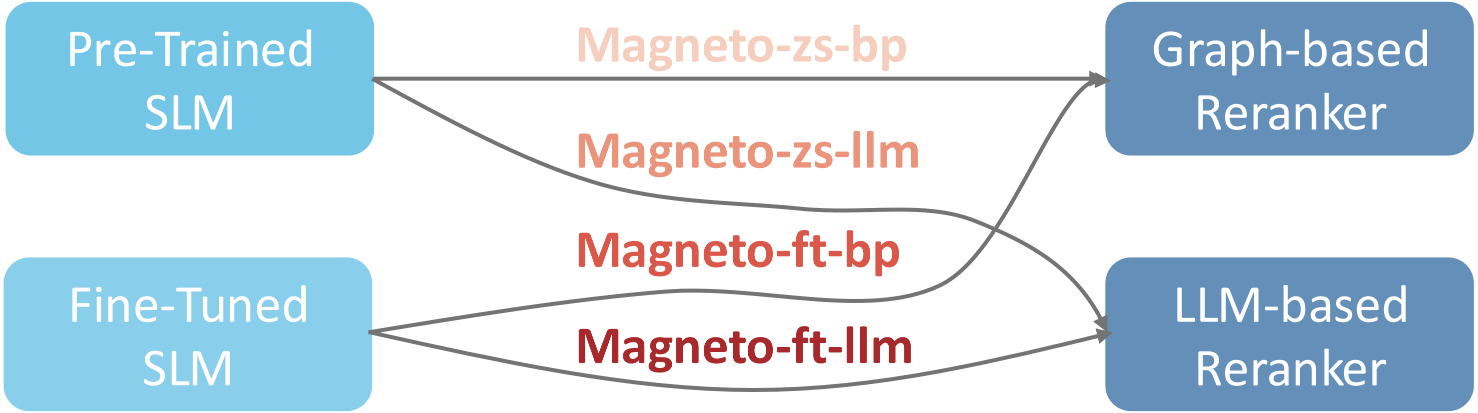
- Given a source column and top-k candidates, LLM **reranks** target columns by generating pairwise matching scores
- One-shot in-context learning ensures consistent scoring

Magneto

- A two-phase framework that combines the advantages of SLMs and LLMs to address their limitations



Magneto Variations



GDC Benchmark

- Based on the Pan Cancer Analysis use case [Li et al. 2023]
- Consists of source datasets produced by **ten studies (16–213 columns each)**
- Mapped to GDC schema: **736 attributes**
- Gold data curated by biomedical researchers from ***NYU Medical School***

Dataset	#Table Pairs	#Cols	#Rows	Match Type
GDC	10	16–736	93–4.5k	Human-Curated
Magellan	7	3–7	0.9k–131k	Human-Curated
WikiData	4	13–20	5.4k–10.8k	Human-Curated
Open Data	180	26–51	11.6k–23k	Fabricated
ChEMBL	180	12–23	7.5k–15k	Fabricated
TPC-DI	180	11–22	7.5k–15k	Fabricated

Access GDC benchmark on Zenodo!

GDC Benchmark

- Captures nuanced semantics
- Scales to large schemas
- Reflects real-world applicability

Complement existing benchmarks!

Dataset	#Table Pairs	#Cols	#Rows	Match Type
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ChEMBL	180	12–23	7.5k–15k	Fabricated
TPC-DI	180	11–22	7.5k–15k	Fabricated

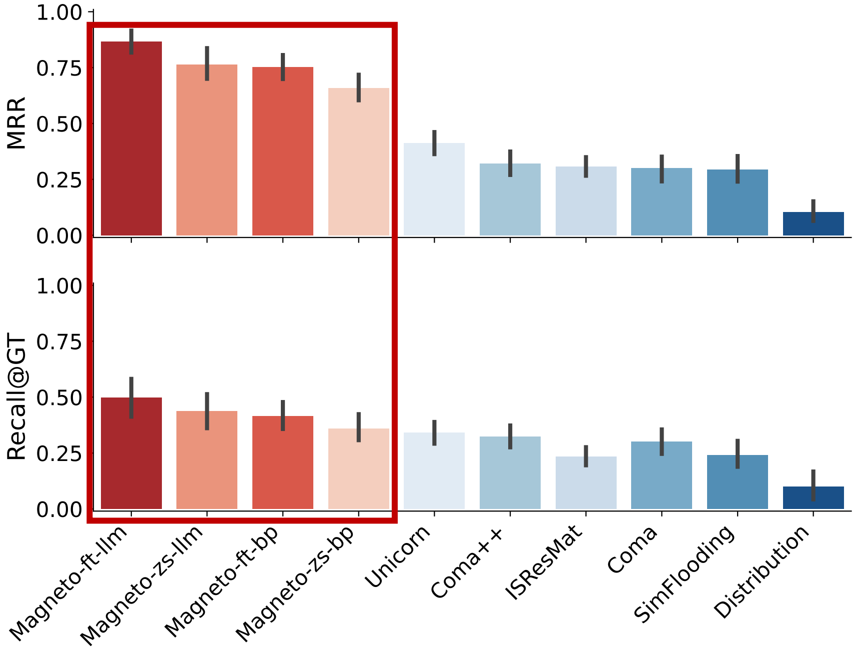
Access GDC benchmark on Zenodo!

End-to-End Accuracy on GDC Benchmark

- Magneto attains higher accuracy on the challenging GDC benchmark than traditional, model-based, and supervised approaches

Rank of the first correct match in the list

How well all correct matches of table S appear at the top of the ranked list

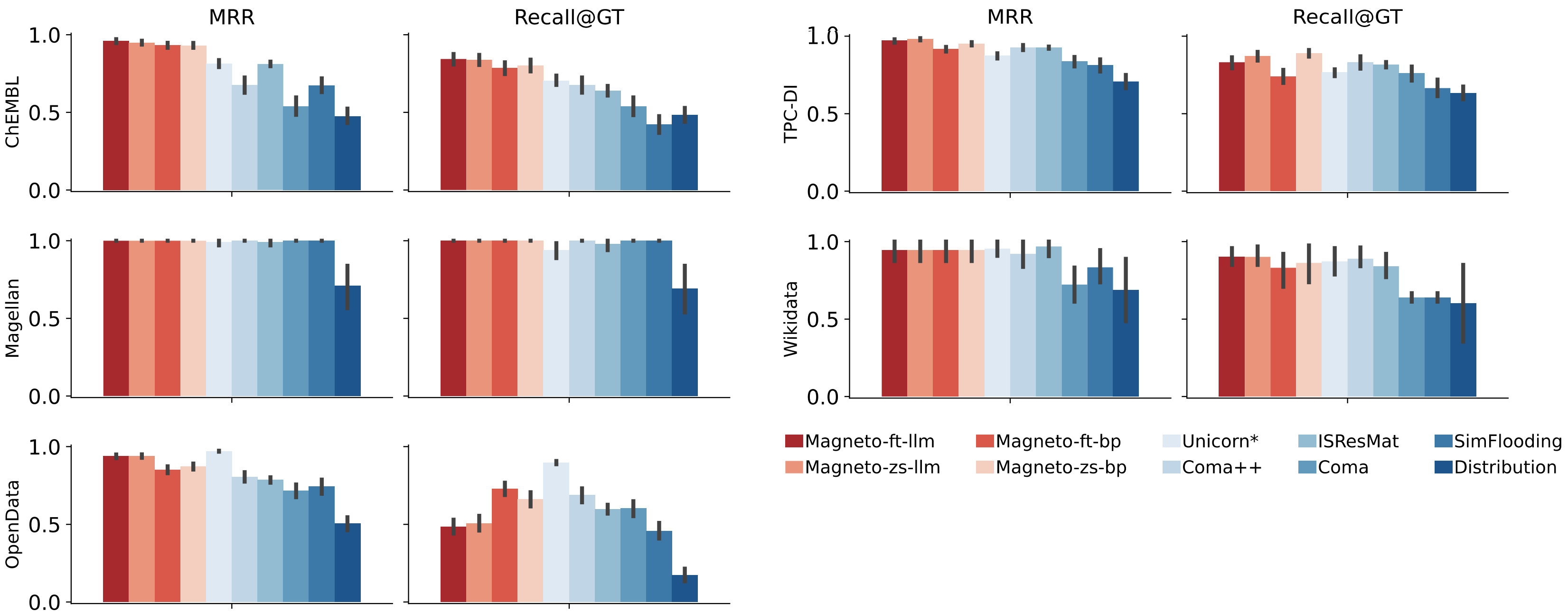


Column-wise view

Table-wise view

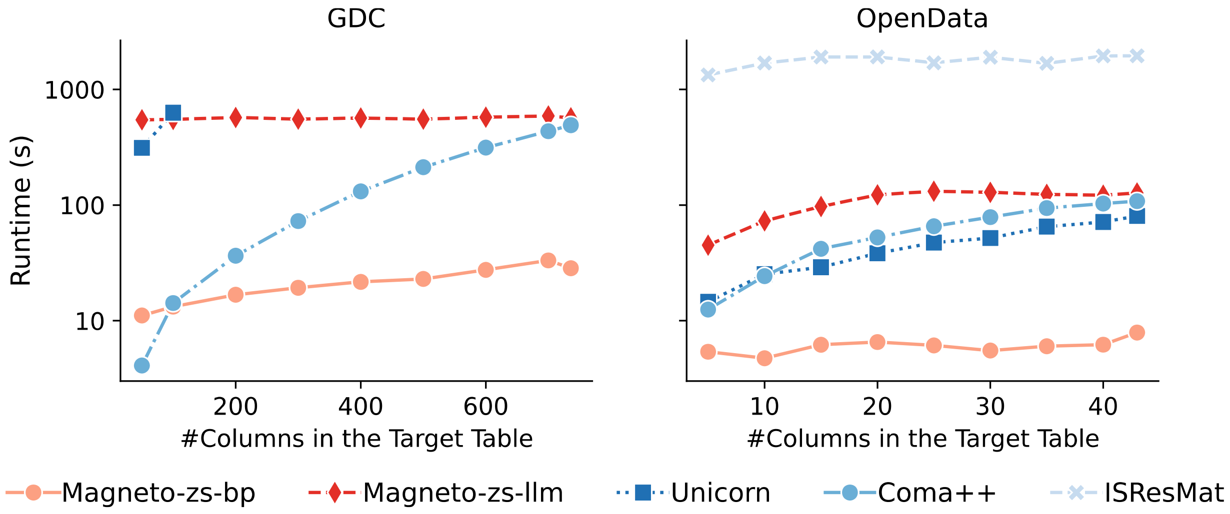
End-to-End Accuracy on Other Benchmarks

- Magneto variants also achieve strong generalization and high accuracy across existing benchmarks [Koutras et al., ICDE 2021]



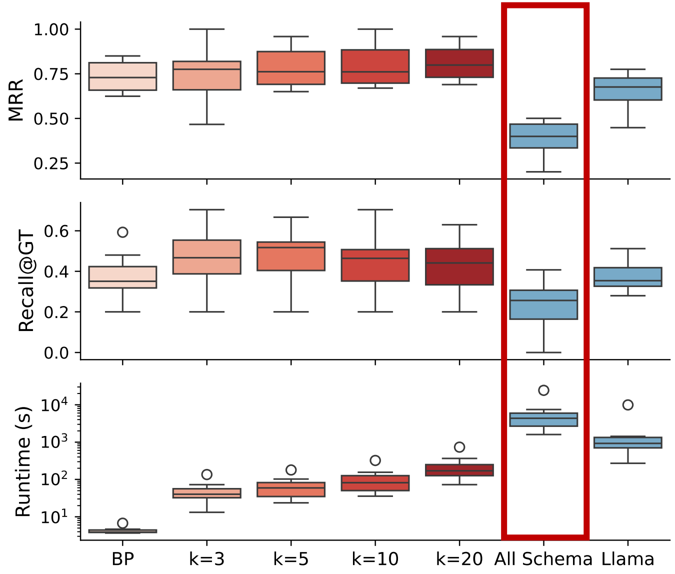
Scalability Assessment

- Magneto-zs-bp is often significantly faster than the other baselines
- Magneto-zs-llm maintains a consistent runtime, completing tasks within minutes, even for datasets with hundreds of columns



Ablations – # candidates sent to LLM

- Trade-offs between MRR, Recall@GT, Cost and efficiency on GDC
- LLM-only approaches are costly and underperform in matching accuracy
- Offer alternatives to balance accuracy and efficiency based on user needs



Ablations – # candidates sent to LLM

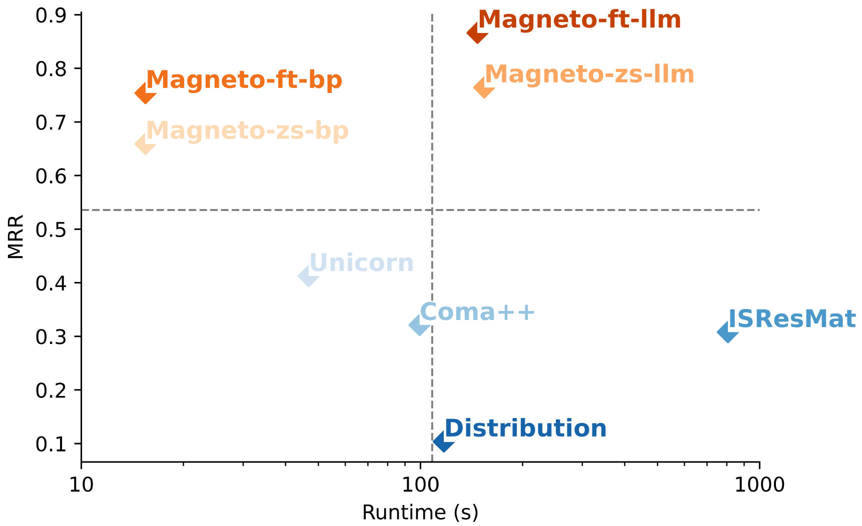
- Trade-offs between MRR, Recall@GT, Cost and efficiency on GDC
- LLM-only approaches are costly and underperform in matching accuracy
- Offer alternatives to balance accuracy and efficiency based on user needs

Check out the paper for more ablations on column serialization, value sampling, synthetic data generation, and backbone choices!

Key Highlight: Magneto is robust across backbone choices, performing consistently on both open- and closed-source models.

Conclusions and Future Work

- Magneto combines SLM efficiency with LLM semantic understanding for cost-effective, accurate matching
- GDC benchmark captures more challenging and semantically complex biomedical integration than existing benchmarks



- Apply specialized post-training to enhance LLM interpretation of structured modalities
- Evolve from static search to dynamic, autonomous agentic workflows

Thanks!

GitHub Repository: https://github.com/VIDA-NYU/magneto-matcher

PyPI Package: https://pypi.org/project/magneto-python/

Contact: yurong.liu@nyu.edu